

- Description

This module manages a smart liquid level monitoring and auto-refill system, ensuring that:

1. **The liquid level in a storage tank is continuously monitored to maintain a stable supply.**
2. **If the liquid level drops below the minimum threshold, the refill system is activated.**
3. **If the liquid level rises above the maximum threshold, the refill system is deactivated to prevent overflow.**
4. **A log of all liquid level readings, refill activations, and shutdowns is maintained for resource management.**

This system helps optimize liquid storage, prevent shortages, and automate refill operations.

- Specification

module SmartLiquidLevelControl

var currentLiquidLevel, maxLiquidLevel, minLiquidLevel: real

var refillActive: boolean

process updateLiquidLevel(changeAmount: real)

functional scenario form:

currentLiquidLevel <= maxLiquidLevel && Liquid level maintained.

||

currentLiquidLevel < minLiquidLevel && refillActive = true, Refill system activated.

||

currentLiquidLevel > maxLiquidLevel && refillActive = false, Refill system deactivated.

end process

end module

- Java Code

```
package code;
class SmartLiquidTank {
    private double maxLiquidLevel;
    private double minLiquidLevel;
    private double currentLiquidLevel;
    private boolean refillActive;
    public SmartLiquidTank(double maxLiquidLevel, double minLiquidLevel
, double initialLiquidLevel) {
        this.maxLiquidLevel = maxLiquidLevel;
        this.minLiquidLevel = minLiquidLevel;
        this.currentLiquidLevel = initialLiquidLevel;
        this.refillActive = false;
    }
}
```

```

    }
    public void updateLiquidLevel(double changeAmount) {
        currentLiquidLevel += changeAmount;
        manageRefillSystem();
        System.out.printf("Current Liquid Level: %.2f liters\n", currentLiquidLevel);
        if (refillActive) {
            System.out.println("Refill System Activated.");
        } else {
            System.out.println("Refill System Deactivated.");
        }
    }
    private void manageRefillSystem() {
        if (currentLiquidLevel < minLiquidLevel) {
            refillActive = true;
        } else if (currentLiquidLevel > maxLiquidLevel) {
            refillActive = false;
        }
    }
}

public class c67_SmartLiquidSystem {
    public static void main(String[] args) {
        SmartLiquidTank tank = new SmartLiquidTank(100.0, 30.0, 50.0);
        // testcase-VT:
        //     tank.updateLiquidLevel(-25.0);
        //     tank.updateLiquidLevel(20.0);
        //     tank.updateLiquidLevel(60.0);
        //     tank.updateLiquidLevel(-10.0);

        // testcase-FT:
        //     tank.updateLiquidLevel(3.139306941570492E305);
        //     tank.updateLiquidLevel(2.563310792133412E305);
        //     tank.updateLiquidLevel(-1.7976931348623157E308);
        //     tank.updateLiquidLevel(-4.092711172425034E306);
        // testcase-Z3:
        //     tank.updateLiquidLevel(-25.0);
        //     tank.updateLiquidLevel(20.0);
        //     tank.updateLiquidLevel(60.0);
        //     tank.updateLiquidLevel(-10.0);

        // testcase-UVT:
        tank.updateLiquidLevel(-25.0);
        tank.updateLiquidLevel(20.0);
        tank.updateLiquidLevel(60.0);
    }
}

```

```

        tank.updateLiquidLevel(-10.0);
    }
}

```

- Coverage report

1. Test case number: 4
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c67_SmartLiquidSystem.java	100.0 %	6	0	6
> c67_SmartLiquidSystem		0	0	0
> SmartLiquidTank	100.0 %	6	0	6

3. Vibration testing instruction coverage: 96.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	86	20,748	20,834
▼ src	0.4 %	86	20,748	20,834
▼ code	0.4 %	86	20,748	20,834
▼ c67_SmartLiquidSystem.java	96.6 %	86	3	89
> c67_SmartLiquidSystem	87.0 %	20	3	23
> SmartLiquidTank	100.0 %	66	0	66

4. Fuzz testing branch coverage: 83.3%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.6 %	5	837	842
▼ src	0.6 %	5	837	842
▼ code	0.6 %	5	837	842
▼ c67_SmartLiquidSystem.java	83.3 %	5	1	6
> c67_SmartLiquidSystem		0	0	0
> SmartLiquidTank	83.3 %	5	1	6

5. Fuzz testing instruction coverage: 96.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	86	20,748	20,834
▼ src	0.4 %	86	20,748	20,834
▼ code	0.4 %	86	20,748	20,834
▼ c67_SmartLiquidSystem.java	96.6 %	86	3	89
> c67_SmartLiquidSystem	87.0 %	20	3	23
> SmartLiquidTank	100.0 %	66	0	66

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c67_SmartLiquidSystem.java	100.0 %	6	0	6
> c67_SmartLiquidSystem		0	0	0
> SmartLiquidTank	100.0 %	6	0	6

7. Z3 instruction coverage: 96.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	86	20,748	20,834
▼ src	0.4 %	86	20,748	20,834
▼ code	0.4 %	86	20,748	20,834
▼ c67_SmartLiquidSystem.java	96.6 %	86	3	89
> c67_SmartLiquidSystem	87.0 %	20	3	23
> SmartLiquidTank	100.0 %	66	0	66

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c67_SmartLiquidSystem.java	100.0 %	6	0	6
> c67_SmartLiquidSystem		0	0	0
> SmartLiquidTank	100.0 %	6	0	6

9. Unimproved V-method instruction coverage: 96.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	86	20,748	20,834
▼ src	0.4 %	86	20,748	20,834
▼ code	0.4 %	86	20,748	20,834
▼ c67_SmartLiquidSystem.java	96.6 %	86	3	89
> c67_SmartLiquidSystem	87.0 %	20	3	23
> SmartLiquidTank	100.0 %	66	0	66